DVD Assignment

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2024102047

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2024102074

1 Part A

1.1 Simulation

```
PASS: 256 vectors checked, 0 errors
tb/tb_mult.v:94: $finish called at 2605000 (1ps)
```

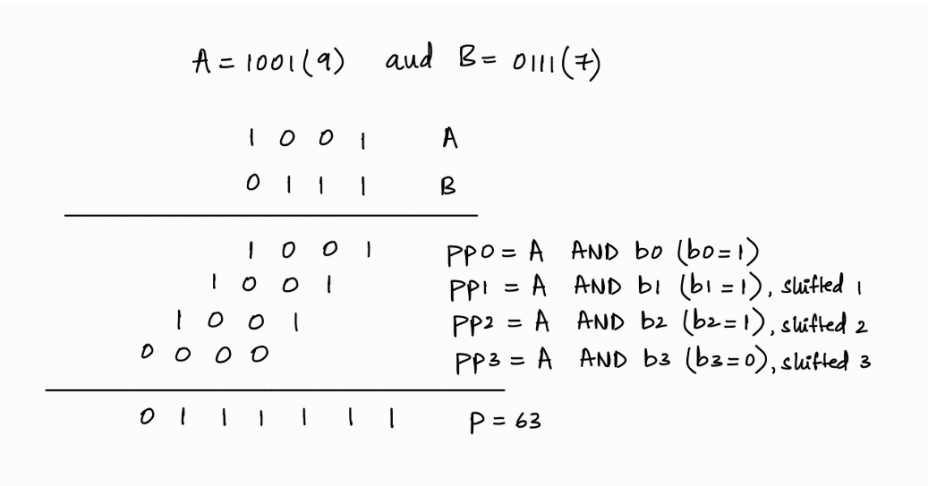
Simulation PASS Line

```
FAIL: 94 errors detected
tb/tb_mult.v:94: $finish called at 2605000 (1ps)
```

Simulation FAIL Line with incorrect Sum Logic

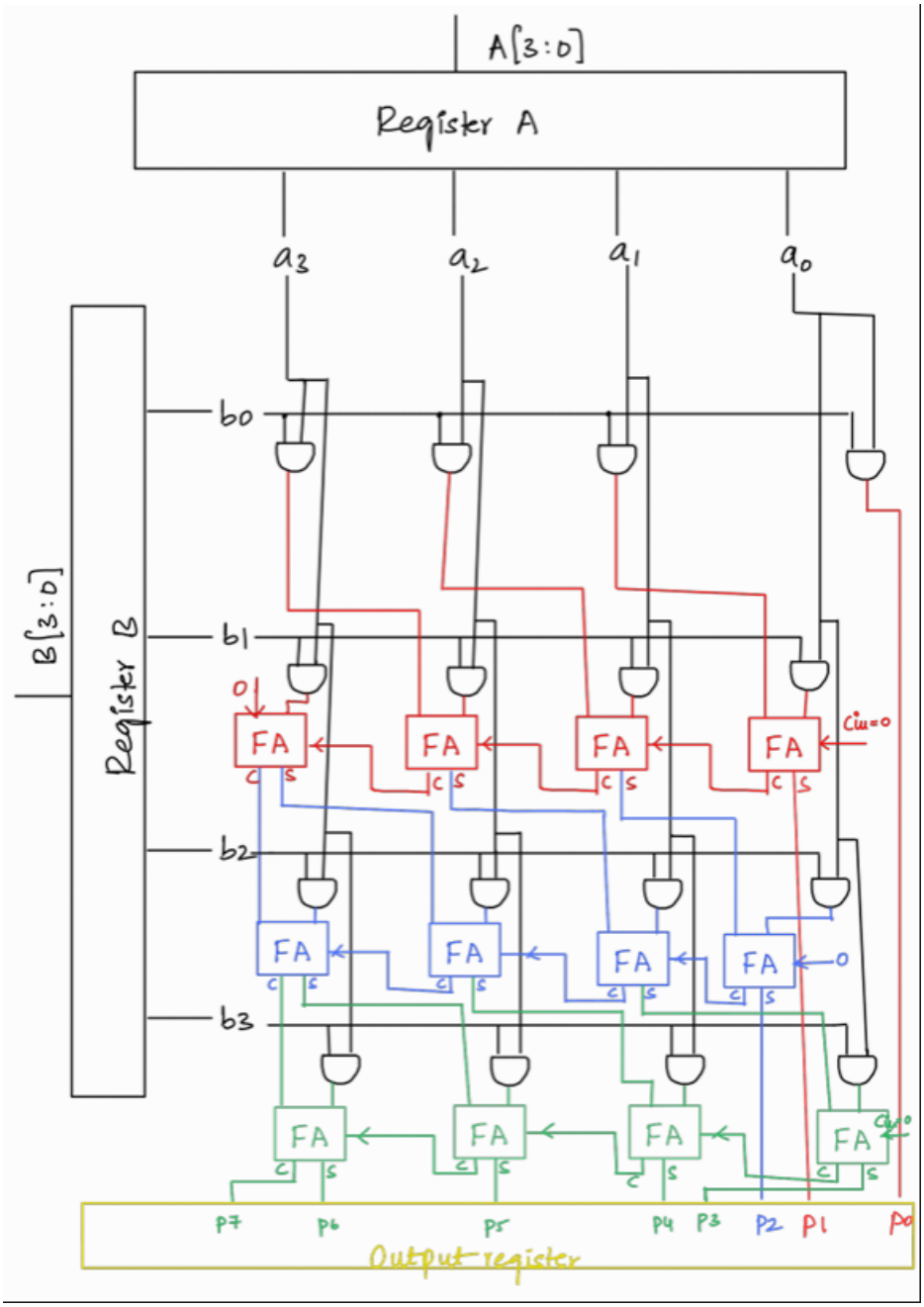
1.2 Hand worked Roll number example

We have considered Sukeerthi's Roll no: 2024102074. So A = 4 + 5 = 9 and B = 7 as instructed. This yields a product of 63.

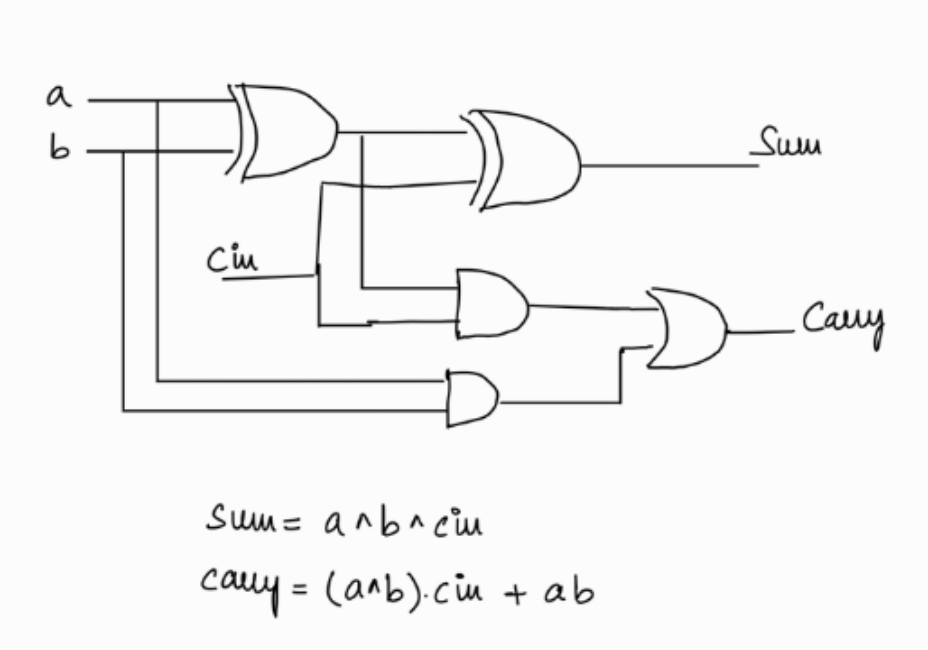


Example considering Roll no. 2024102074

1.3 Block Diagram



Block Diagram



Full Adder Circuit

1.4 Waveform



GTK Wave with marked latency for a clock period of 10ns (A = 9 and B = 7)

2 Part B

2.1 Results Table

Metric	**Value**
Total number of cells	67
Number of flip-flops	16
Total cell area (μm^2)	784.50
Flip-flop area (total) (μm^2)	400.38
Individual flip-flop area (μm^2)	25.02
Flip-flop area (% of total)	51.04%
The 3 most-used cell types and their counts	`dfrtp_1` (16), `nand2_1` (9), `xnor2_1` (7)
Netlist simulation result (PASS/FAIL line)	PASS: 256 vectors checked, 0 errors

Table 1: Logic Synthesis Results from Yosys Statistics Report

2.2 Simulation

```
PASS: 256 vectors checked, 0 errors
tb/tb_mult.v:94: $finish called at 2605000 (1ps)
```

Netlist PASS Line

3 Part C

3.1 Results Table

Metric	Value
Worst slack at $T = 10$ ns	7.020 ns
T_{min} (ns)	2.980 ns
F_{max} (MHz)	335.57 MHz
Slack at $T = T_{\text{min}}$ (ns)	0
Critical path start point	_106_
Critical path end point	_100_
Number of cells on the critical path	10
Latency (ns)	5.960 ns
Throughput (million results per second)	335.57

Table 2: Static Timing Analysis Results from OpenSTA

```
  0.411    0.411 v _106_/Q (sky130_fd_sc_hd__dfrtp_1)
  0.218    0.629 v _048_/X (sky130_fd_sc_hd__and4_1)
  0.428    1.057 ^ _050_/Y (sky130_fd_sc_hd__a2111oi_0)
  0.094    1.151 v _051_/Y (sky130_fd_sc_hd__a21oi_1)
  0.218    1.369 ^ _053_/Y (sky130_fd_sc_hd__mux2i_1)
  0.374    1.742 v _069_/X (sky130_fd_sc_hd__xnor3_1)
  0.371    2.113 v _074_/X (sky130_fd_sc_hd__maj3_1)
  0.376    2.489 v _076_/X (sky130_fd_sc_hd__maj3_1)
  0.355    2.844 v _080_/X (sky130_fd_sc_hd__xor3_1)
  0.000    2.844 v _100_/D (sky130_fd_sc_hd__dfrtp_1)
           2.844   data arrival time
```

List of Cells along Critical Path as obtained from mult_array_timing.rpt

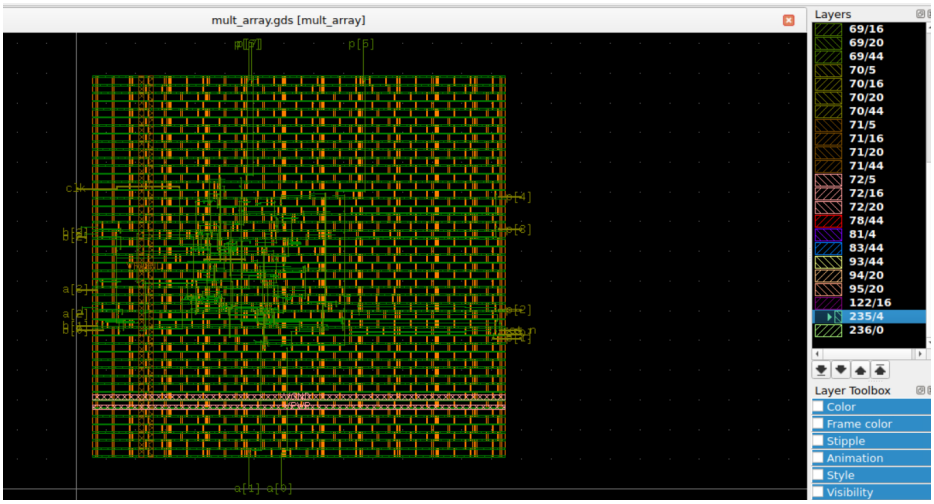
Out of the 10 cells in the critical path, 2 are flip-flops and the remaining 8 are combinational logic gates. This aligns closely with the theoretical estimate of $2N = 2(4) = 8$ logic stages for a 4-bit multiplier.

4 Part D

4.1 Results Table

Metric	Value
CLOCK_PERIOD used (ns)	7.0
Number of runs needed	7 attempts (verify run history)
Core area (μm^2)	17759.5
Cells after synthesis (from Part B)	67
Cells after place and route	1901 total instances
Kinds of cells added	Buffers, fill cells, tap cells, decap cells
Wirelength (μm)	2348
Setup slack (ns)	0.089
Post-route $F_{\max}$ (MHz)	144.70
$F_{\max}$ from Part C (MHz)	335.57
DRC errors	0
LVS errors	0

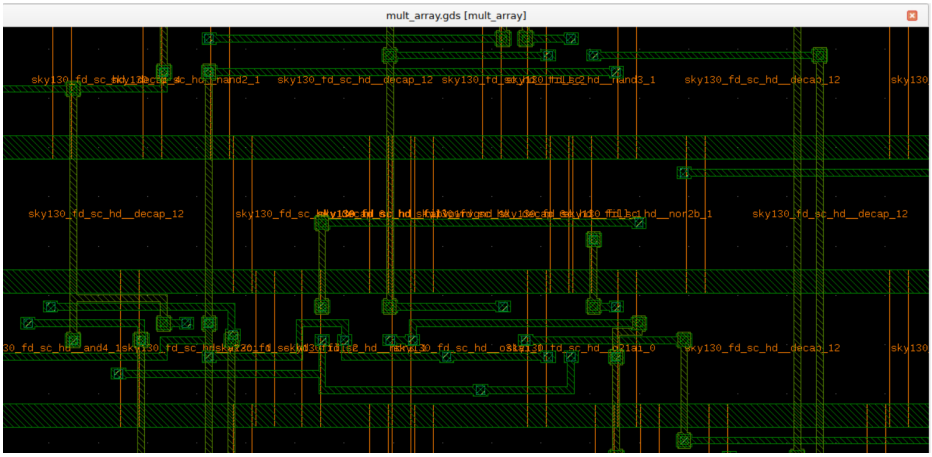
Table 3: Physical Design Results from LibreLane



Top Level View Layout



Hierarchial View



Zoomed view showing standard cell rows and metal wires.

5 Answers for Part 10

5.1

Synthesis produced 51 logic cells, calculated by subtracting the 16 flip-flops from the total of 67 cells. The theoretical design requires 16 AND gates to generate the partial products and 12 full adders to perform the addition. However, the final number of cells is not simply $16 + 12 = 28$, because these are conceptual hardware components rather than the actual number of standard cells used after synthesis. Yosys converts the Verilog design into a logic network, and ABC optimizes this network and maps it to the available SKY130 standard-cell library. Each full adder may require multiple logic cells, while Boolean logic optimization can combine operations, share logic, and use complex gates such as AOI, OAI, and majority gates. Therefore, the final implementation contains 51 logic cells and 16 flip-flops, giving 67 total cells.

5.2

The critical path identified by OpenSTA starts at register `_106_` and ends at register `_100_`. According to the timing report, the path has 10 cell instances, consisting of 2 flip-flops and 8 combinational logic cells. The combinational cells along the path are an AND4 gate, an A2111OI gate, an A21OI gate, a MUX2I gate, an XNOR3 gate, two majority (MAJ3) gates, and an XOR3 gate. At a clock period of 10 ns, the data arrival time is 2.844 ns, and the worst negative slack (WNS) is 7.020 ns, indicating that the path meets the timing requirement. For a 4-bit multiplier, the approximate critical-path estimate is $2N = 2 \times 4 = 8$ full-adder delays. The 8 combinational cells in the measured path are broadly comparable to this estimate, but they are not equivalent to eight full adders. Synthesis optimizes the arithmetic logic and maps it into different standard cells, each with its own propagation delay. Therefore, the actual critical-path delay depends on the types and arrangement of the mapped cells, rather than simply the number of full adders. The measured path delay of 2.844 ns is consistent with a short, optimized combinational path in the 4-bit multiplier.

5.3

The total cell area after synthesis is 784.5024 μm^2, of which the 16 flip-flops occupy 400.384 μm^2. Therefore, the percentage of the total cell area occupied by flip-flops is calculated as:

$$\text{Percentage of area occupied by flip}-\text{flops} = \frac{400.384}{784.5024} \times 100 = 51.04\%$$

Thus, flip-flops occupy approximately 51.04% of the total cell area. This percentage is large because the multiplier is a small, pipelined design that uses registers to store the input operands and the output product. The 16 flip-flops account for nearly half of the total area, while the remaining area is occupied by combinational logic cells that perform partial-product generation and addition. Flip-flops also require more silicon area than many simple logic gates because they contain multiple internal transistors to store data and maintain their state. Therefore, in a small design with a limited number of logic gates, the area contribution of the registers becomes proportionally large.

5.4

The post-route maximum operating frequency ($F_{\max}$) is lower than the $F_{\max}$ obtained in Part C because Part C performs static timing analysis on the synthesized netlist, whereas the physical-design flow considers the effects of placement, routing, and interconnect parasitics. The resistance and capacitance of the routed wires introduce additional delays, and the physical-design flow may add cells that also affect the timing and area of the design. In our Part C results, the minimum clock period was 2.980 ns, giving an $F_{\max}$ of 335.57 MHz. In Part D, using a clock period of 7.0 ns and a reported setup slack of 0.089 ns, the post-route $F_{\max}$ is calculated as

$$F_{\max} = \frac{1000}{7.0 - 0.089} = 144.70MHz$$

The physical-design flow adds several types of cells for implementation and reliability. Buffers strengthen signals and help drive large loads or long wires. Fill cells fill unused spaces in standard-cell rows to maintain layout continuity and meet physical-design requirements. Tap cells connect the substrate and well regions to the appropriate power supplies, helping prevent latch-up. Decap cells provide local charge storage to reduce power-supply noise. These additional cells and the routed interconnects increase the physical implementation's area and can affect its timing compared with the synthesized design.

5.5

Design Rule Check (DRC) verifies whether the physical layout follows the manufacturing rules specified by the semiconductor process. It checks geometric constraints such as minimum wire width, minimum spacing between metal wires, layer overlaps, and other layout requirements to ensure that the chip can

be manufactured correctly. Layout Versus Schematic (LVS) verifies whether the physical layout matches the intended circuit netlist by comparing the components, their connections, and the overall circuit structure. In our LibreLane flow, both DRC and LVS checks passed with zero reported errors, indicating that the layout satisfies the checked manufacturing rules and matches the intended circuit connectivity. Both checks are essential before manufacturing because a layout that violates design rules may cause fabrication defects, while a layout that does not match the netlist may produce a chip that functions incorrectly.

5.6

For an N-bit array multiplier, the number of full adders can be estimated using the formula $N(N-2)+N-1$, which simplifies to N^2-N-1. For a 4-bit multiplier, the estimated number is $4^2-4-1=11$, approximately 12 full adders as specified in our assignment. For an 8-bit multiplier, the estimated number is $8^2-8-1=55$ full adders. Therefore, increasing the multiplier width from 4 bits to 8 bits increases the number of full adders from approximately 12 to 55, which is about 4.6 times more. The critical-path delay can be estimated as proportional to $2N$ full-adder delays. For a 4-bit multiplier, this is $2\times 4=8$ full-adder delays, while for an 8-bit multiplier, it becomes $2\times 8=16$ full-adder delays. Using the measured Part C critical-path delay of 2.844 ns as the 4-bit baseline, the estimated 8-bit critical-path delay is $2.844\times\frac{16}{8}=5.688$ ns. Thus, doubling the multiplier width approximately doubles the critical-path delay, while the number of full adders increases by roughly four times.

$$FullAdders = N^2 - N - 1$$

$$N = 8: \quad 8^2 - 8 - 1 = 64 - 8 - 1 = 55$$

Critical-Path Estimate: $\mathrm{D}_8 = D_4 \times \frac{2(8)}{2(4)} = 2.844 \times \frac{16}{8} = 5.688ns$

6 Repository Link

Repository Link: https://github.com/su-keerthi/DVD-Assignment

7 Contributions

Part A (Functional Verification & Testbenches) and Part B (RTL Synthesis & Area/Power Analysis) were implemented by Sukeerthi Kattamuri. Part C (Place and Route (P&R), DRC/LVS Checks) and Part D (Timing Analysis, Sweeps, & KLayout Inspection) were implemented by Padmanjali Nethi. Report documentation and final review were handled jointly by Sukeerthi Kattamuri and Padmanjali Nethi.